



Physics Worksheet

Electric Charges and Fields

Name: _____ Student #: _____ Student Email #: _____

1. How does a lightning rod work?
2. Show that the Coulomb Force is conservative
3. How much heavier would a proton have to be to counteract the repulsive field it exerts on a positive charge with its gravitational field?

4. Imagine an electric $\vec{E}$ field pointing East and a dipole submerged within it.

(a) Draw a diagram of this, where there is a net torque on the dipole

(b) Draw the two equilibrium states the dipole can be in. Label which is stable, and which is unstable

(c) Which of the diagrams above has the highest, and the lowest potential energy